

Primer

The avian brain

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There is a reason this primer is called ‘The avian brain’ and not ‘The birdbrain’ even though the latter term might be more accessible. The term birdbrain was long used as a derogative, yet I hope that this primer will inspire you to see it more as I do — as a compliment. This primer will take you on a short tour. First, I will explore superficial differences between birds and mammals. Then I will briefly mention the behavioral capabilities of birds and explain why the bird brain has the neural makeup to generate these behaviors. This will lead me to briefly describe some key neural circuits and structures and to conclude with the ongoing challenge to identify functional differences in the neural substrates of avian and mammalian cognition. The aim of this short article, therefore, is not to give a full review of the avian brain but rather to highlight important aspects with a focus on the comparison to the mammalian brain and cognition.

The avian brain is no birdbrain

Why the term ‘birdbrain’ is a derogative is almost self-evident, even without going into the history of the term itself. Birds are often small animals with small absolute brain sizes. This is most striking in the extremes. The largest mammalian brain, that of the blue whale (*Balaenoptera musculus*), can weigh an impressive 9 kg while the largest brain of any bird, that of the emperor penguin (*Aptenodytes forsteri*) only weighs 44 g. To make matters look worse for the bird brain, birds are the surviving dinosaurs and the evolutionary lines of birds and mammals split over 320 million years ago. Consequently, the brains of birds look nothing like you would expect from a smart mammal. In mammals, the seat of higher cognition is the beautifully layered cortex that forms the outer shell of the telencephalon. In larger mammals we can observe the typical gyrfication of the cortex, the magnification of the surface relative to the rest of the brain. Since birds lack the separation between grey and white

matter, their brains have a smooth surface (except for the cerebellum) and the outer shell of the telencephalon does not show the histologically visible layers of the cortex. These gross anatomical differences have long led to the belief that birds never fully evolved a cortex but that much of the avian brain instead consisted of a massively hypertrophied striatum.

This idea of a considerably less evolved bird brain stands in stark contrast to the impressive cognitive abilities and, if you allow, to the evolutionary success of birds. Most places on earth are inhabited by birds and birds are often the most noticeable animals in our cities, be that because loud birdsong wakes you up in the morning or because you’re surrounded by pigeons blatantly scavenging for food while you enjoy your espresso. Mammalian scavengers such as rats or mice will typically act much stealthier around humans. This great evolutionary success comes with great diversity. Bird species have specialized for all sorts of evolutionary niches and evolved a great variety of appearances and behaviors. Central to this article are species that have emphasized the evolution of cognition, which is most prominent in parrots and in corvids (ravens, crows, jays, jackdaws etc.). But the workhorses of

cognitive neuroscience, pigeons and finches, should also not be forgotten. Pigeons were long a favourite model in psychology, giving us a wealth of behavioural data and inspiring countless models. Pigeons were also an instrumental model for the advancement of our understanding of the neural basis underlying avian cognition. The fascinating work in songbirds gave neuroscience a unique and in-depth perspective on the neural networks underlying vocal behaviour and vocal learning. It is safe to say that, besides mammals, birds are the second evolutionary line that gave rise to highly sophisticated behavior and cognition. In fact, birds show many of the cognitive abilities that are often thought of as typical only for primates, such as tool use, mirror self-recognition, theory of mind and sensory consciousness to name but a few examples.

Why are birds capable of such feats even though their brains are superficially so much smaller and different from mammalian brains? Let me first revisit the issue of brain size (Figure 1). Birds are generally small and light-weight animals with correspondingly small brains. The brain of a rhesus macaque (*Macaca mulatta*), one of the most established models in cognitive neuroscience,

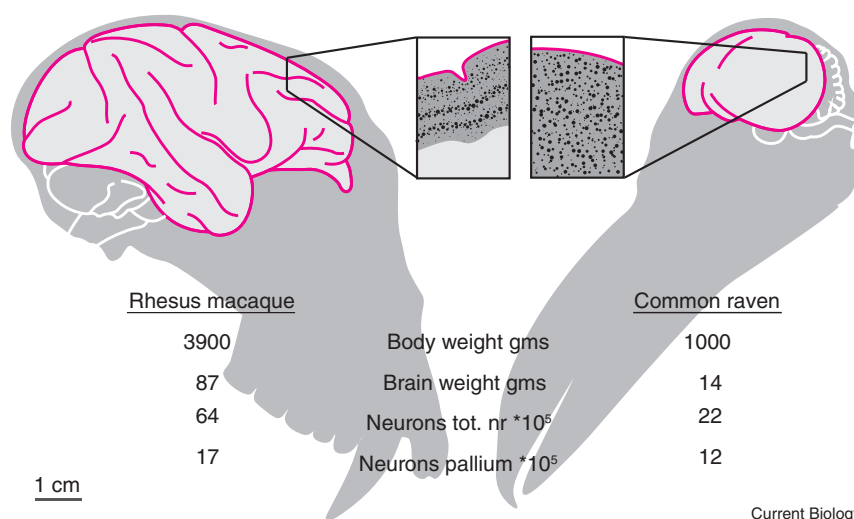


Figure 1. Comparing mammalian and avian brains.

Brain and skull (grey background) of a rhesus macaque (*Macaca mulatta*) and a common raven (*Corvus corax*). Superficial differences between mammalian and avian brains such as size and gyrfication are immediately visible. The insets illustrate the presence of a layered cortex in the primate and its absence in the bird. The table highlights large differences in brain weight between the species. Importantly, it shows how the difference diminishes when comparing the total number of neurons and even further when comparing pallial neuron numbers between the species.

weighs 87 g. That is over six times the weight (14 g) of the brain of a raven (*Corvus corax*), a textbook example of a smart bird. While not without its own shortcomings, a common way to conceptualize brain size is not in absolute terms but relative to body weight. Proportionally to the heavy rhesus (3.9 kg), its brain weighs 2.2% while in the much lighter raven (1 kg) the brain weight makes up 1.4% of the body weight. But perhaps the most telling comparison is offered by the unit of neural computation, the number of neurons. And here the macaque comes out on top — but not by as much as one may expect. The brain of a macaque contains about three times the number of neurons as that of the raven (~64 vs ~22 × 10¹⁵ neurons). This also indicates that, compared to most other species, certainly considering body weight, ravens have a remarkably high number of neurons.

Comparing absolute number of neurons between the species is relevant, yet considering where in the brain these neurons are located makes it much more meaningful. In fact, this is where history got the bird brain very wrong. Contrary to previous ideas, the avian brain is not mostly a hypertrophied striatum. Instead, we now know that birds, much like mammals, have a large pallium. (In mammals, the pallium consists of cortex, hippocampus and parts of the amygdala.) If we further compare the number of neurons in the pallium of the macaque to that in the pallium of the raven, the difference between the species becomes much less prominent (17 × 10¹⁵ vs 12 × 10¹⁵ neurons). It goes without saying that the normalization — the division of neuron number by body size — would put the raven far ahead of the macaque (but absolute neuron number is computationally likely more relevant).

These numbers also have another implication. Ravens have proportionally more neurons in the pallial structures than in the rest of the brain and this ratio is higher than it is in the macaque. The same holds if we compare ravens to bird species that are less associated with higher cognition, such as chickens. This highlights the degree of enlargement that corvids and parrots show in parts of the brain that are associated with cognitive ability,

an observation that is in line with an evolutionary specialization to become cognitive generalists and allowing especially corvids to gain a foothold in many diverse ecological niches around the world.

Let me assume that you're convinced that, from a purely quantitative perspective, bird brains might have the same potential as mammalian brains. As mentioned in the first paragraph, superficially the brains of birds and mammals look vastly different. Birds don't show the compartmentalization between grey and white matter and, consequentially, telencephalic neurons are not arranged into the highly structured outer layer that constitutes the mammalian cortex. The differences in organization between the species have often made a direct one-to-one mapping of different parts of the cortex to different parts of the avian brain difficult. The avian pallium is homologous to the mammalian pallium and it consists of two distinct regions — the wulst (German word for bulge) at the dorsal surface and the dorsal ventricular ridge (DVR), which is further subdivided into meso-, nido- and arcopallium. But instead of focusing on the homologies between avian and mammalian structures I will now assume a functional, networks-driven perspective.

Avian telencephalic networks

A look at the 'connectome' offers a good way to frame this approach. Recent work found that the brain of pigeons, like mammals, has a small-world network consisting of sensory, limbic, premotor and associative modules together with a structural core. This central core is made up of extensively connected parahippocampal, premotor and prefrontal hubs, a network architecture that is virtually identical in mammals. This is a remarkable finding given the evolutionary distance and known structural differences between the species such as the absence of the corpus callosum in the avian brain.

At the network level the organizational similarity continues, for instance in the visual system. Vision is the dominant sense in birds and many avian species have excellent vision with high spatial and temporal resolution and precise colour vision that extends into the ultraviolet

spectrum. Much like in mammals there are two main visual projections, a thalamofugal (collothalamic) and a tectofugal (lemnthalamic) system (Figure 2). While the thalamofugal system is dominant in mammals, this is not the case in all birds. In some avian species, mostly with forward-facing eyes such as owls, the thalamofugal system is dominant while in other species, such as pigeons, the tectofugal system is dominant. The sheer size of the pigeon tectum highlights its importance for vision (Figure 2). In addition to these two central visual systems the centrifugal system is also well developed in birds. It is a feedback system to the retina that is only weakly present in mammals and entirely absent in humans.

Birds can easily be considered experts in prototypical hippocampal functions such as spatial and episodic memory and navigation, rendering the hippocampus an obvious region of interest. However, the avian hippocampus has long been a somewhat elusive structure. On one hand stood a sizable body of largely lesion and inactivation studies arguing for similar functions to the mammalian counterpart. On the other hand stood considerable anatomical differences and difficulties to firmly establish neurophysiological similarities to the mammalian counterpart. Recently, however, place- and head-direction cells have clearly been identified in the avian hippocampus, putting the notion of functional similarity on much firmer ground.

A canonical circuit for cognition

The similarity to mammals does not stop at the visual streams. While birds show neither the separation between grey and white matter nor the layered cortex, recent results indicate a strong resemblance to the cortex at the circuit level (Figure 2). Three lines of evidence point in this direction. Initially, genetically identified neuron types linked individual cortical layers to neural clusters in the avian brain. Very recently modern histological methods finally made it possible to identify the connectivity between comparably unstructured and three dimensionally arranged clusters in the avian brain. This work revealed a basic circuit that

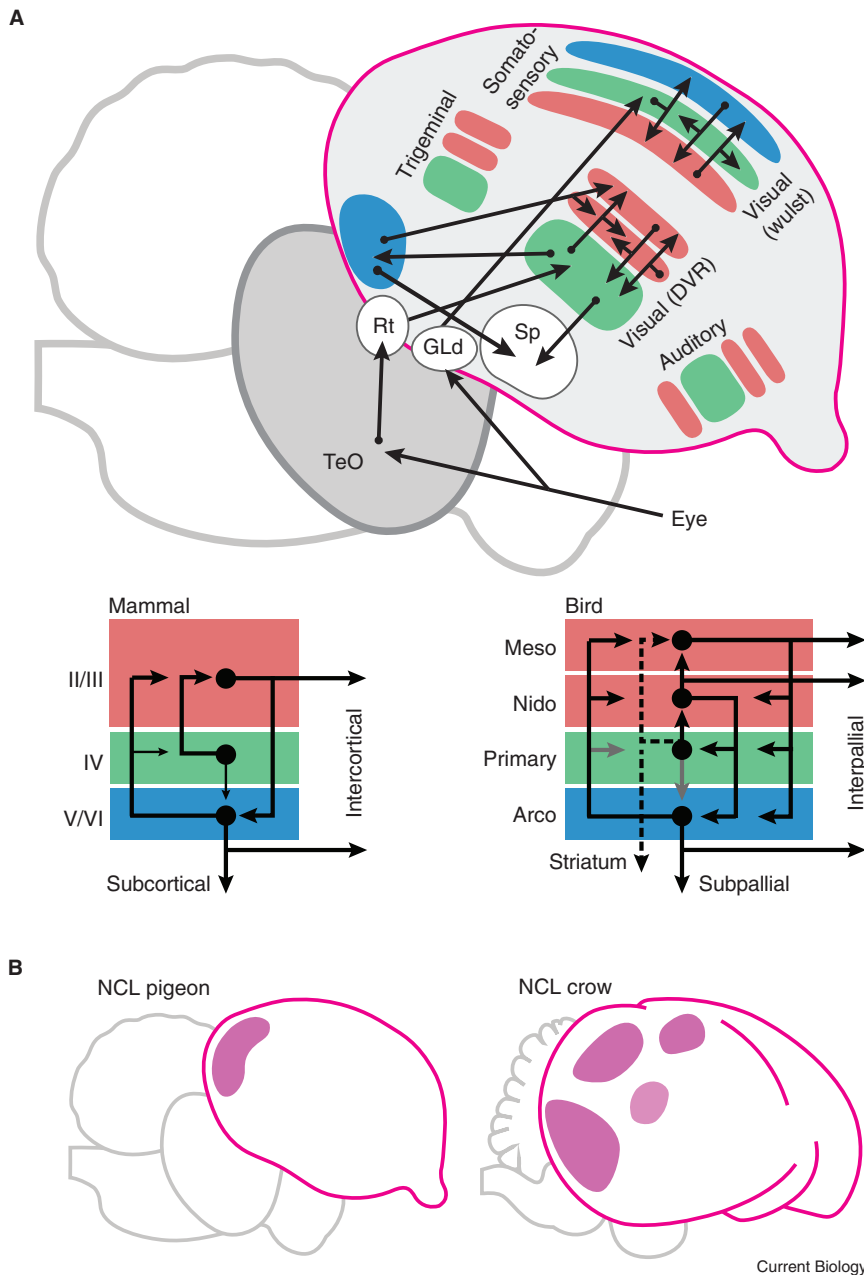


Figure 2. Avian sensory networks and ‘prefrontal’ structures.

(A) Schematic depiction of avian sensory pallial circuits visualized in a pigeon brain. Thalamo- and tectofugal projections arrive in two distinct visual areas, in the wulst and the dorsal ventricular ridge (DVR), respectively. The circuit is repeated in an almost identical fashion in the two visual and in all other primary projections. The side-by-side schemata depict the canonical mammalian cortical and avian sensory circuits (here visual tectofugal). Grey arrows indicate projections present only in the wulst, dashed lines indicate projections only present in the DVR. (B) Diversification of the nidopallium caudolaterale (NCL). In pigeons (left) the NCL is confined to a single region. New data suggest that NCL diversified into several distinct regions in crows and other songbirds. Rt: nucleus rotundus, TeO: optic tectum, GLd: nucleus geniculatus lateralis pars dorsalis, SP: subpallial areas (striatum, thalamus, brainstem), meso: mesopallium, nido: nidopallium, arco: arcopallium, primary: primary sensory structure.

is remarkably similar within all sensory domains and shows a repetitive pattern of horizontal and vertical connections.

Lastly, further evidence for this circuit was found in the auditory system of finches using single neuron recordings.

This circuit is not only largely comparable between different sensory modalities within and between different bird species (pigeons and owls), it also closely resembles the canonical mammalian circuit, the cortical column. Even though birds do not have a cortex, at the circuit level, the sensory structures of birds and mammals might therefore be more similar than expected with neural processing relying on a comparable — but not identical — canonical circuit.

The avian ‘prefrontal cortex’

In mammals the entire cortex is comprised of this canonical circuit, admittedly with some small variations between areas. But the picture is somewhat different in birds. While the circuit is present in all sensory modalities it is absent in the highest associative structures. In birds, the functional counterpart of the mammalian prefrontal cortex (PFC) is the nidopallium caudolaterale (NCL). The NCL can be identified as the part of the avian pallium that receives the highest dopaminergic innervation and here dopaminergic fibers form a distinct morphological feature. In these so called ‘baskets’ fibers wrap around individual perikarya indicating a tight dopaminergic control of the underlying neuron. It has been speculated that this morphology thereby implements similar principles as layer-specific dopamine release in the cortex.

In the last decades a multitude of techniques including lesions, pharmacological intervention and neurophysiological recordings were used to examine the NCL. To keep it brief, one of the main and reoccurring findings is that the similarities between NCL and PFC seem to outweigh the differences. To name but a few examples, neural activity in NCL correlates with rewards, rules, categories, numerosity and information held in working memory. Even the neural mechanisms limiting working memory capacity are highly comparable between the two structures and we have recently shown that neural oscillations of local field potentials also resemble those in the PFC. Thus the convergent evolution of prefrontal functions between birds and mammals resulted in similar neural correlates, implying limited degrees

of freedom when it comes to the underlying neural computations in spite of different anatomical substrates.

Recent work has provided a new perspective on the cognitive differences between bird species. The NCL is relatively larger in species with higher cognitive ability than in the less fortunate species. Pigeons, for instance, have a relatively larger NCL than chickens and this increase can also be observed from non-corvid songbirds (e.g. zebra finches) to corvid songbirds such as carrion crows. But what makes things interesting is that the NCL in pigeons and chickens is confined to a single location while in songbirds the NCL has diversified into multiple structures. This leaves room for functional comparative research not only between species with different NCL structures but also in species with similar structure and different relative NCL sizes. To date, no functional work has been done to investigate the differences between these architectures or their different roles in an executive network.

To summarize the central points, the great evolutionary distance between birds and mammals gave rise to a number of striking differences in neural organization, the most notable of which is the lack of a separation between grey and white matter in birds. Such differences must not necessarily imply a disadvantage in birds. For instance, the higher neuron density in the smaller avian brain implies shorter distances between connected neurons. Considering that shorter distances can imply higher processing speeds, the fact that the brain of the rhesus macaque is over six times the size of the brain of the raven might make the cognitive abilities of the raven less surprising. Recent years have shown that the more we learn about the avian brain the more we discover similarities to mammals, be that at the circuit level or in the neural correlates of cognitive processes. For instance, while the cortical organization of the avian brain is not (immediately) obvious, closer examination uncovered similar blueprints of cognition in both mammals and in birds. But the discovery of what is comparable also gives us grounds to discover the true differences. The

question why the avian equivalent of the cortical column is only found in sensory areas but not in the NCL might point to an important difference between birds and mammals and indicate a fundamentally different neuroanatomical solution that still gives rise to very similar neural processing. A case of such an entirely different neural circuitry could in turn challenge our models of cortical processing. But to get to that point we need more systematic research on the neural networks that underlie avian cognition.

DECLARATION OF INTERESTS

The author declares no competing interests.

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Primer

Avian vision

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The primary reliance of birds on vision is readily and frequently asserted. Casual observations of birds completing their everyday behaviours is sufficient to convince most observers that birds are using vision to control their key behaviours. This assertion is supported by evidence that in most bird species relatively large portions of their brains are devoted to the analysis of information from vision. Also, the so-called intelligent behaviours of birds seem to be based primarily upon visual information. Thus, gaining information from vision, and using it to guide sophisticated behaviours, is seen as a vital component of what makes birds so fascinating.

Well-founded comparisons between the vision of different bird species are limited to a relative handful of the 11,000+ bird species currently described. Even among this handful of species, however, there is good evidence for wide differences in their vision. Each species appears to employ a unique suite of visual information that guides them in the execution of their particular daily tasks in different light environments; from pecking to lunging at food items, from bright sunlight to woodlands at night, from flying in open skies to diving below water. In this primer, I will give an overview of bird vision, its underlying physiology and anatomy, and the key factors that have driven its diversity

The drivers of bird vision

The key drivers in the evolution of bird senses are generally thought to be the informational demands of foraging — the detection, acquisition, and screening of food items — and the detection of predators. However, the visual tasks of foraging and predator detection make antagonist demands for information, and these can be seen played out in the vision of most birds. The informational demands for controlling flight are best considered to be met within the requirements set by the primary sensory challenges of foraging and predator detection.

